

Evaluating and Reducing Hallucinations in LLM-Based Medical Report Summarization

Progress Report, Fall 2026 (COMP 698C)

Srilaya Ponangi, M.S. Computer Science

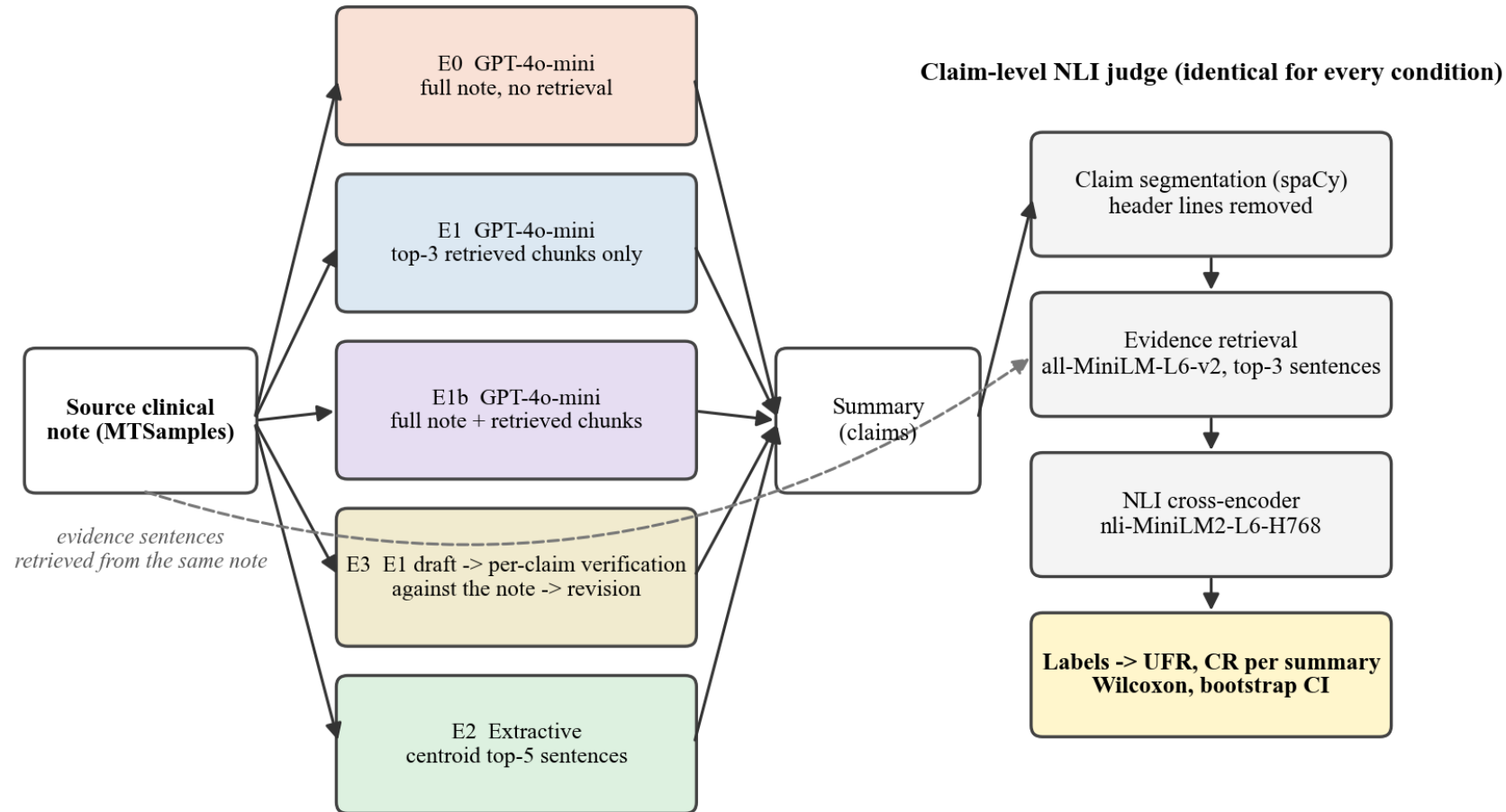
Advisor: Dr. Taehyung (George) Wang

California State University, Northridge, September 14, 2026

Project and research questions

- Goal: measure hallucinations in LLM patient-facing summaries at the claim level and test whether retrieval grounding reduces them
- RQ1: how often are GPT-4o-mini summary statements unsupported by or contradicting the note (claim-level NLI judge)?
- RQ2: which statements are unsupported, and how does the judge itself err against medical experts?
- RQ3: does document-grounded RAG reduce unsupported and contradicted statements, versus an extractive bound, and at what coverage cost?
- Five conditions compared on 50 MTSamples notes: E0 zero-context LLM, E1 RAG (excerpts only), E1b RAG (note + excerpts), E3 RAG + Chain-of-Verification, E2 centroid extractive
- Judge: spaCy claims, all-MiniLM-L6-v2 retrieval (top-3), cross-encoder/nli-MiniLM2-L6-H768; metrics UFR and CR per summary

Pipeline (identical judge for every condition)



Every summary is scored by the same claim-level NLI judge; evidence is retrieved from the same note.

Data status

- MTSamples (public): 4,999 transcriptions, 624 eligible consult and discharge notes, 50 sampled (seed 42); 2,016 claims labeled
- MIMIC-IV-Note v2.2 (PhysioNet, credentialed): downloaded May 14, 2026, integrity verified; staged for a future local-model re-run, not sent to any API
- ann-pt-summ v1.0.1 (PhysioNet, DUA signed): 210 expert-annotated patient summaries, 423 unsupported-fact spans; used to validate the judge with local models
- The PhysioNet archive download was incomplete (934 MB of >3.4 GB); the 14 annotation files were extracted and verified against the SHA-256 manifest
- Ethics: CITI 'Data or Specimens Only Research' and 'Conflicts of Interest' completed April 10, 2026; no credentialed text leaves the laptop

Corrections since the May 2026 report

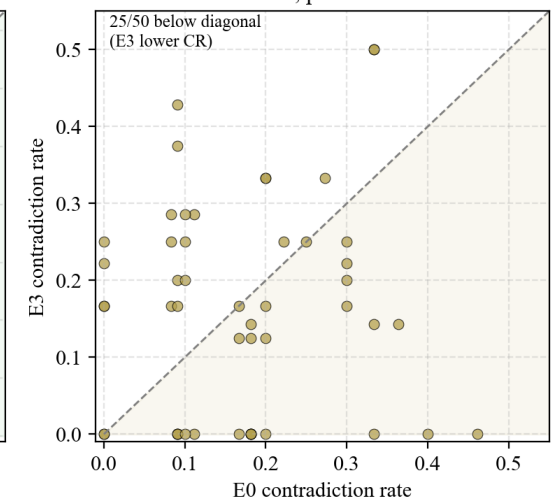
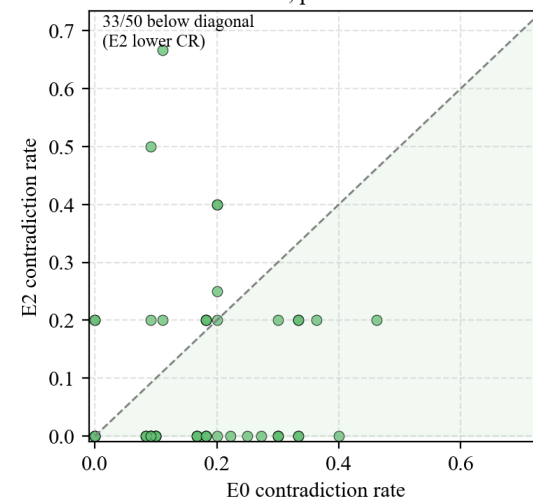
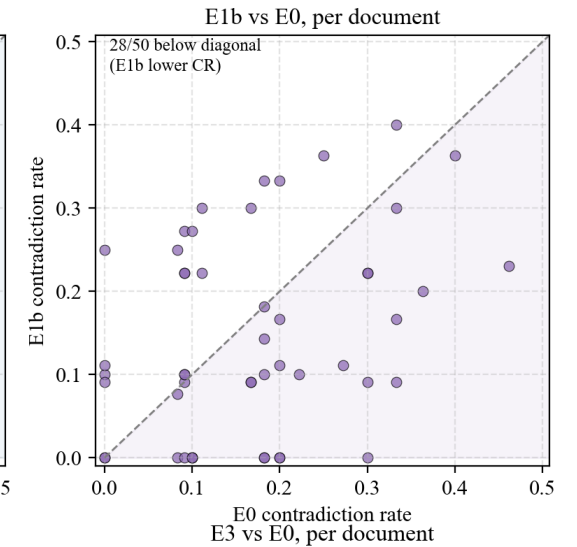
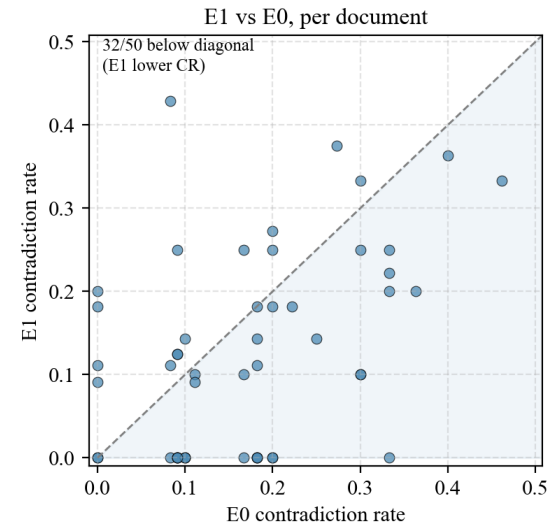
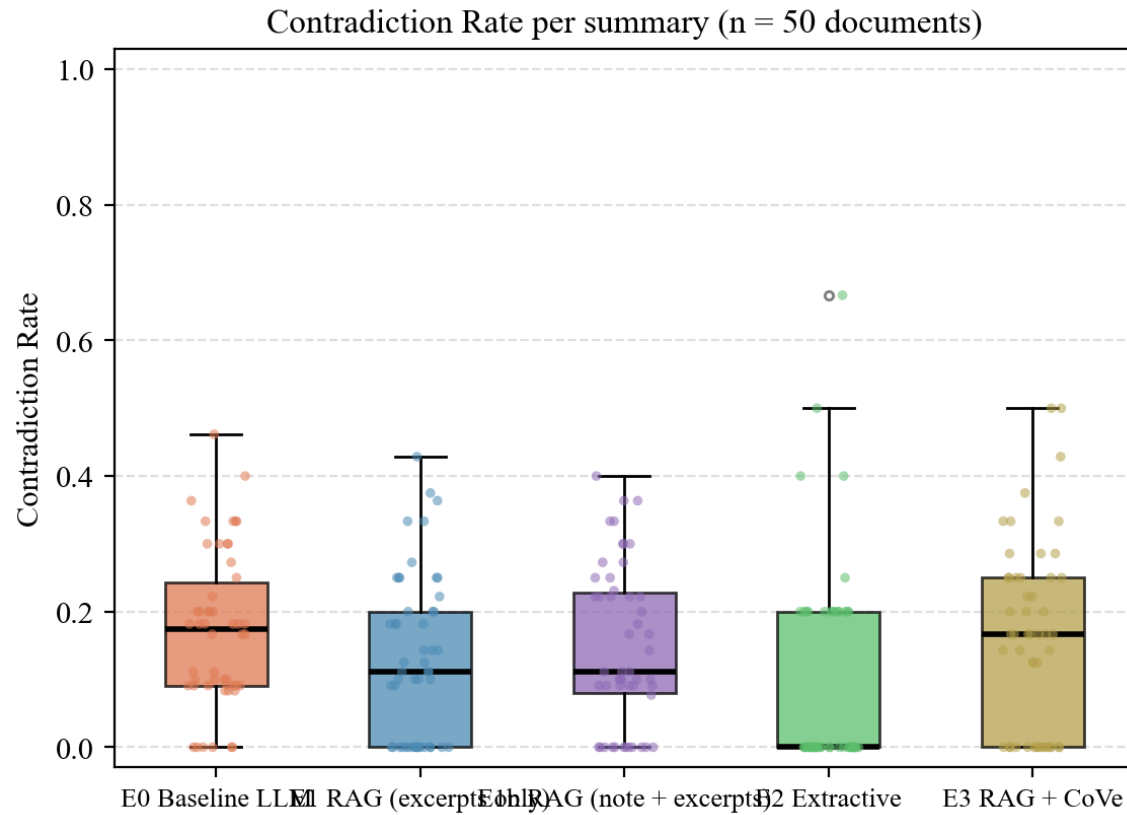
Issue found	Correction	Effect
Markdown headers ('**Key Findings:**') counted as claims	Header filter; 131 E0 and 31 E1 header lines removed	E1 vs E0 CR effect: -0.073 ($p < 0.001$) $\rightarrow$ -0.042 ($p = 0.011$)
E1 described as 'note + excerpts', overlapping chunks, structured query	Description corrected: excerpts only; non-overlapping 5-sentence chunks; query = description + first 300 chars	Adds E1b (note + excerpts) as an implemented extension
NLI rule described as argmax over a concatenated premise	Corrected: max over 3 separate pairs, 0.5 thresholds	Aggregation variants now tested against experts
'48 percentage-point' median CR drop; Xie et al. mis-citation; APA/IEEE mix	Arithmetic fixed; citation replaced (Asgari 2025); IEEE with DOIs	64-entry reference list
requirements.txt missing 4 packages; 'bootstrap CIs' claimed but absent	Packages added; bootstrap CIs, effect sizes, Holm adjustment implemented	bash run.sh --offline reproduces every number

Main results (n = 50 documents; header lines and abstentions excluded)

Metric	E0 zero-context	E1 RAG excerpts	E1b RAG + note	E3 RAG + CoVe	E2 extractive
UFR mean	0.811	0.807	0.759	0.688	0.122
CR mean	0.172	0.130	0.148	0.164	0.096
Supported share of claims	19%	20%	24%	33%	89%
Claims per summary	10.4	8.9	10.2	5.9	4.8
Words per summary	182	156	180	120	69
Coverage of source sentences ($\cos \geq 0.6$)	16.7%	13.4%	18.5%	16.6%	18.3%

Paired Wilcoxon versus E0: E1 CR $p = 0.011$ (−24%), E1 UFR $p = 0.977$; E1b UFR $p = 0.020$; E3 UFR $p < 0.001$ (−15%; versus E1 $p < 0.001$), E3 CR $p = 0.913$; E2 UFR $p < 0.001$. Verification lowers unsupported facts the most, by deleting claims and abstaining; retrieval alone lowers contradictions modestly; the extractive bound has a non-zero judge error floor.

Contradiction rate: distributions and per-document pairs

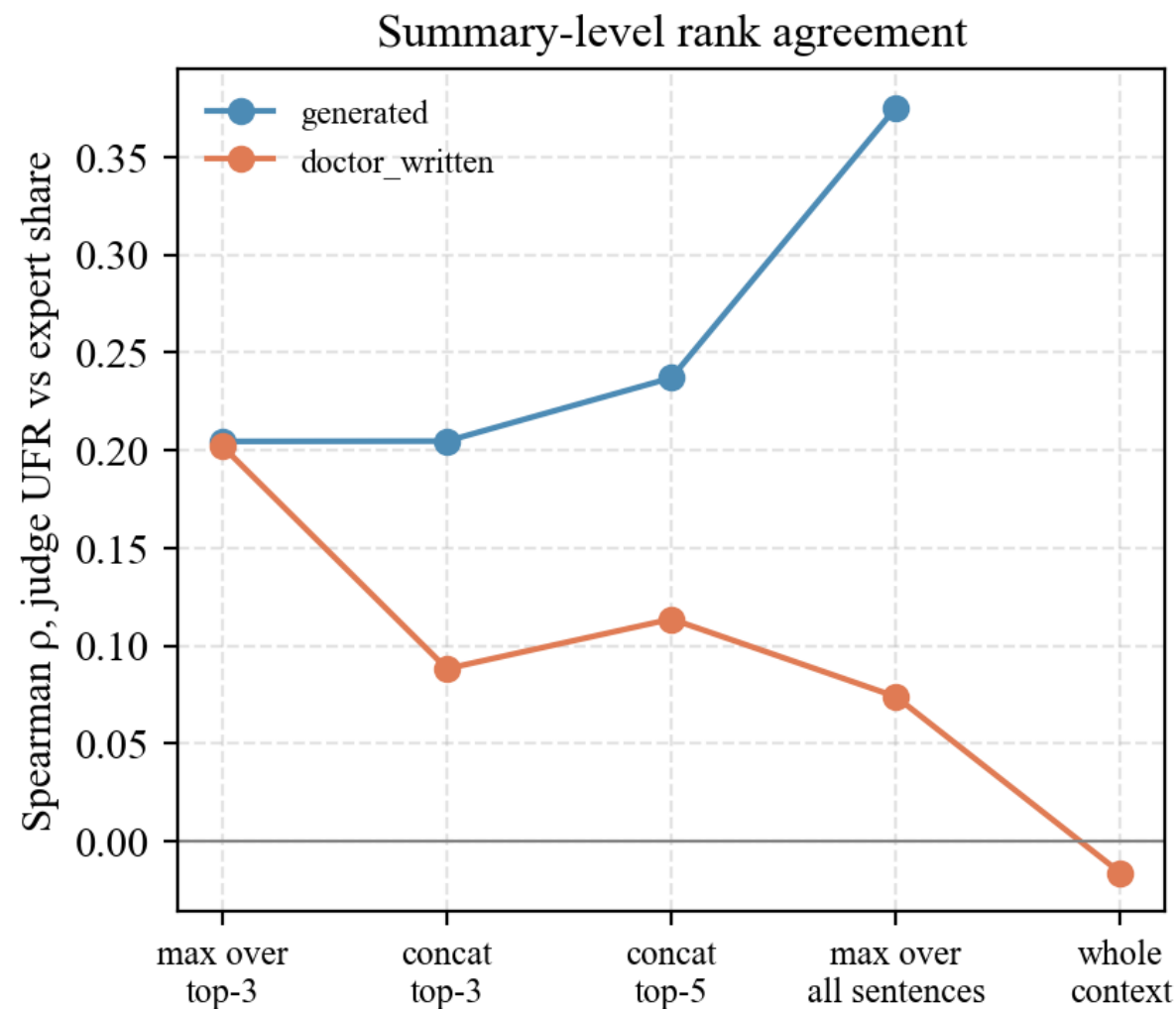
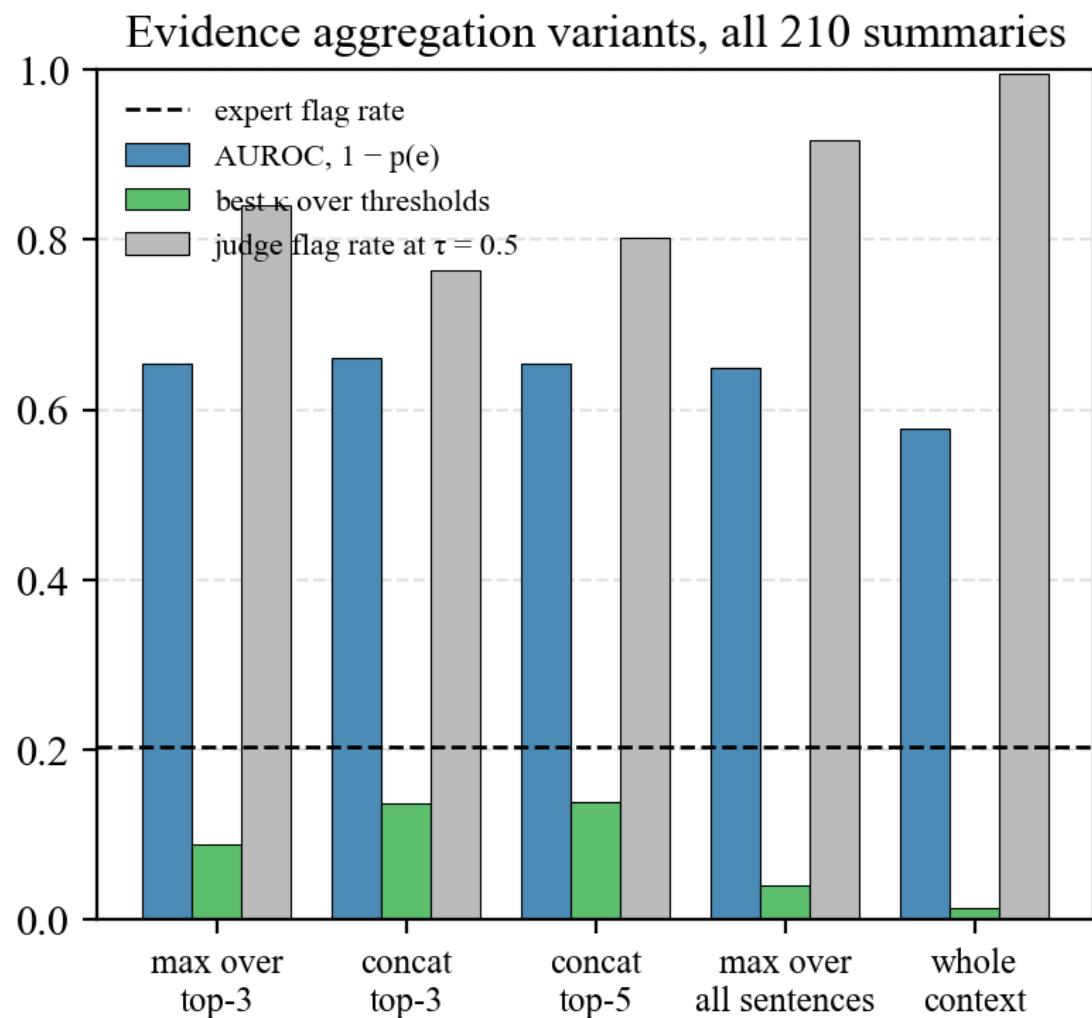


Headline finding: the judge disagrees with medical experts

Group	Sentences	Expert-flagged	Judge-flagged (UFR)	Precision	Recall	Kappa	AUROC
all	1,781	20.4%	84.1%	0.23	0.94	0.06	0.65
generated	805	12.5%	83.7%	0.14	0.96	0.04	0.68
doctor written	976	26.8%	84.3%	0.30	0.94	0.07	0.66
llama 70b original	159	26.4%	83.0%	0.30	0.95	0.10	0.67
gpt4 zero shot	151	6.0%	87.4%	0.06	0.89	0.00	0.50

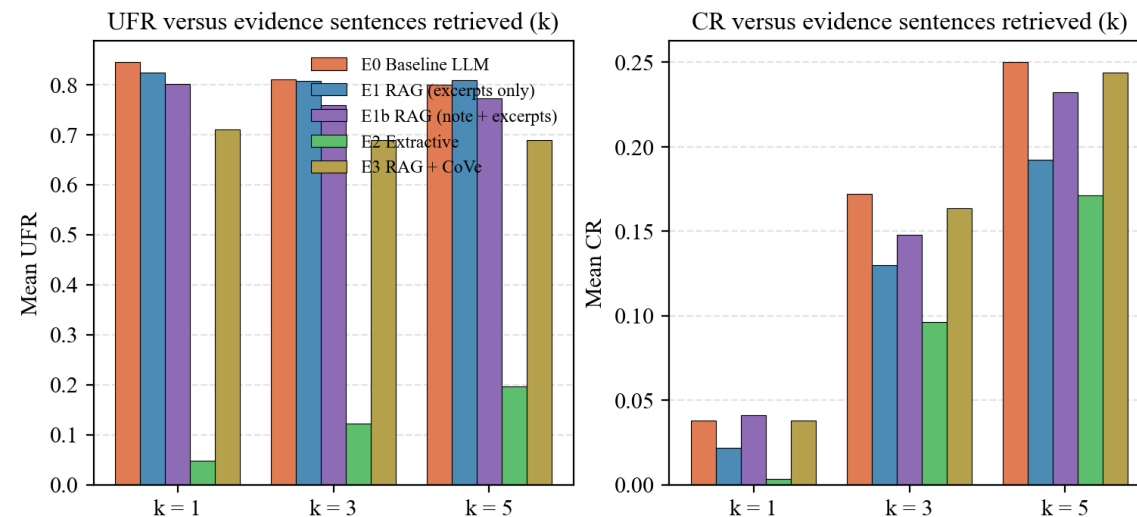
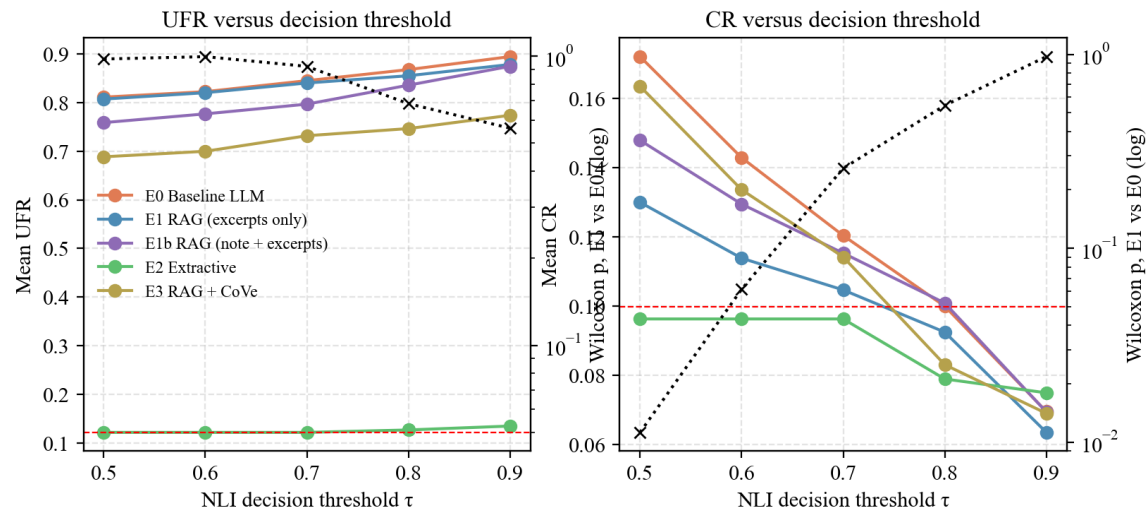
On 1,781 expert-annotated sentences the judge flags 84% where experts flag 20%; kappa 0.06 is chance level, and the judge ranks systems in nearly the reverse order of the experts. Absolute UFR/CR are not hallucination rates.

No aggregation rule rescues the judge

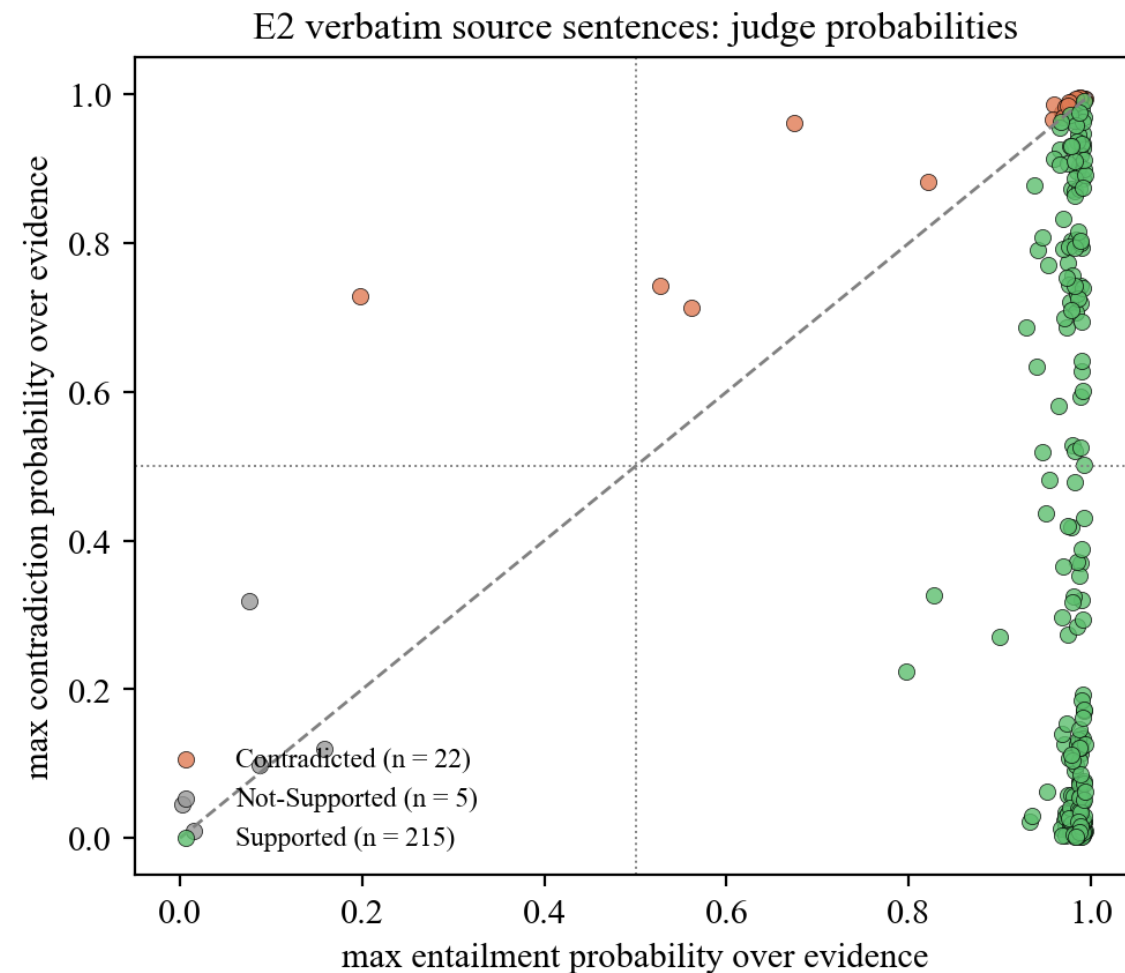
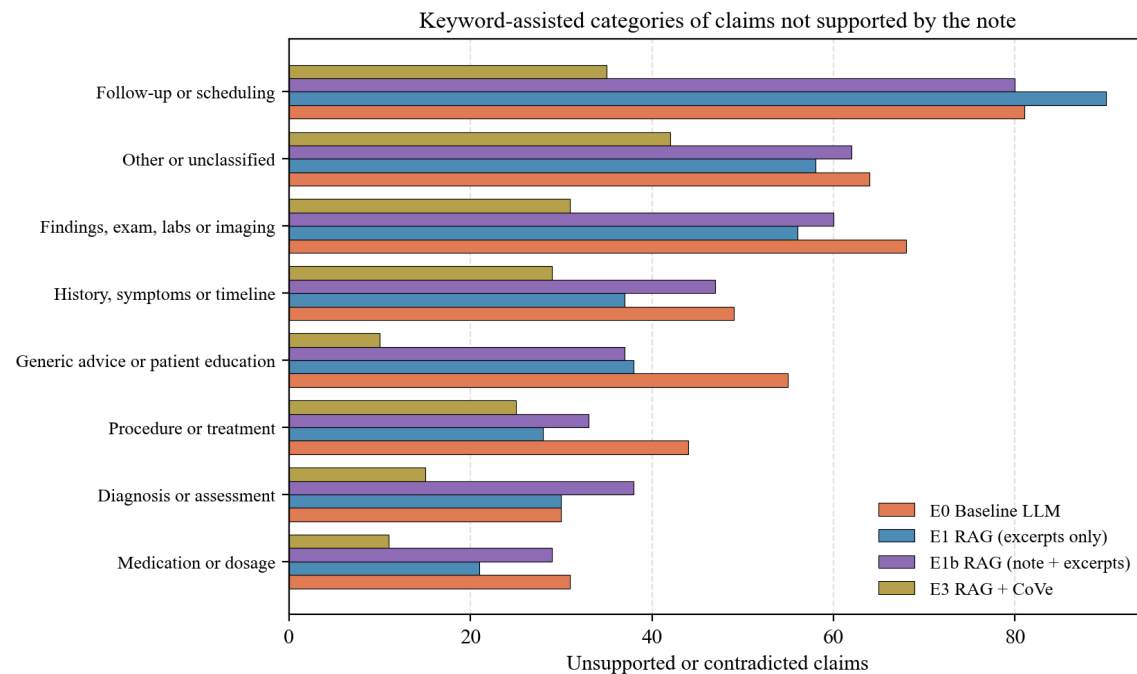


Best kappa over all variants and thresholds: 0.14; best AUROC 0.66. The limiting factor is the small general-domain NLI model on paraphrased patient-facing sentences.

Robustness: the RAG effect holds only at the default threshold



What remains unsupported, and the judge's own errors



Thesis writing status

- Complete draft generated from the result files: 107 pages in CSUN format (1-inch margins, 12-pt Times New Roman, double-spaced, roman-numbered preliminary pages, arabic body)
- Chapters: 1 Introduction, 2 Literature Review, 3 Data and Preprocessing, 4 Methodology, 5 Results, 6 Discussion, 7 Conclusion and Future Work; 64 IEEE references with DOIs; Appendices A to E (prompts, per-document results, worked examples, repository, threshold sweep)
- New content since May: two new conditions (E1b, E3), header-filter correction, judge validation against expert annotations, five aggregation variants, threshold/top-k/cleaned-evidence ablations, coverage proxy, negation analysis, error taxonomy, ethics and governance section
- Open items: committee member names on the signature page; advisor review; title alignment with the ETD planning form (question answering removed from scope)
- Every table and figure is regenerated by one command; the document rebuilds in about two minutes

Plan to the defense (aligned with CSUN ETD deadlines)

By	Milestone
Sep 19, 2026	Advisor review of this complete draft; committee names confirmed
Sep 30, 2026	Sampling-variance runs (several generations per note) if API budget allows; optional MIMIC re-run with a local open-weight generator
Oct 9, 2026	ETD Planning Form filed (title without question answering)
Oct 16, 2026	Revised draft to advisor for approval; Canvas upload and Turnitin check after approval
Oct 30, 2026	Draft to committee (two-week review); Doodle poll with 1-hour slots over 3 to 4 weeks
Nov 6, 2026	ETD preliminary format review
Nov 16 to 24, 2026	Defense (40 min presentation, 10 min Q&A); slides to advisor one week before
Dec 4, 2026	Final ETD upload after revisions

The December 12 defense date in the May report is after the December 4 ETD deadline and has been moved.

Risks and requests

- Judge validity: the central instrument agrees with experts at chance level; the thesis reports this as its main methodological finding and proposes stronger judges as future work
- MIMIC generation: requires a zero-data-retention agreement or a local open-weight model; kept optional
- Schedule: two weeks of committee review plus advisor approval must precede the Nov 6 format review
- Request to advisor: confirm committee members, review the complete draft, and advise on the title change